

Notes for ‘Riemann sums and definite integrals’

Important Ideas and Useful Facts:

- (i) Riemann sums, areas under curves and the definite integral:

Suppose that $a < b$ and consider a continuous function f defined for x along the interval $a \leq x \leq b$. Divide this interval into n subintervals (not necessarily of equal width) by choosing

$$t_0 = a < t_1 < t_2 < t_3 < \dots < t_{n-1} < t_n = b,$$

so that $[t_{i-1}, t_i]$ denotes the i th subinterval for $i = 1$ to n . Choose points

$$x_1, x_2, \dots, x_n$$

within successive subintervals, that is, $x_i \in [t_{i-1}, t_i]$ for $i = 1$ to n . Denote by Δx_i the width of the i th subinterval, that is

$$\Delta x_i = t_i - t_{i-1},$$

for $i = 1$ to n . We think of Δx_i as a small change in x , as x moves across the interval from t_{i-1} to t_i . Then the (signed) area under the curve (that is, bounded by the curve, the x -axis and the vertical lines $x = a$ and $x = b$) is approximated by the *Riemann sum*

$$\sum_{i=1}^n f(x_i) \Delta x_i = f(x_1) \Delta x_1 + f(x_2) \Delta x_2 + \dots + f(x_n) \Delta x_n.$$

Each summand $f(x_i) \Delta x_i$ is the area of an approximating rectangle of width Δx_i and height $f(x_i)$. The exact value of the (signed) area under the curve is the limit of the Riemann sums

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i,$$

and called the *definite integral of f from a to b* . This limit exists regardless of the choices made in partitioning the interval, or in choosing each x_i in the i th subinterval, for $i = 1$ to n , provided the widths of the subintervals approach zero as $n \rightarrow \infty$.

- (ii) **Upper and lower Riemann sums:** A Riemann sum is called *lower* if the rectangles always sit beneath the curve and it provides a lower bound for the area under the curve.

If the curve is increasing then the lower Riemann sum is formed always by choosing the left-hand endpoints of the subintervals.

If the curve is decreasing then the lower Riemann sum is formed always by choosing the right-hand endpoints of the subintervals.

A Riemann sum is called *upper* if the rectangles always sit above the curve, and it provides an upper bound for the area under the curve.

If the curve is increasing then the upper Riemann sum is formed always by choosing the right-hand endpoints of the subintervals.

If the curve is decreasing then the upper Riemann sum is formed always by choosing the left-hand endpoints of the subintervals.

(iii) Properties of the Definite Integral:

- (a) $\int_a^b c \, dx = c(b-a)$ and $\int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx$ for any constant c .
- (b) $\int_a^b f(x) \pm g(x) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$.
- (c) $\int_a^a f(x) \, dx = 0$ and $\int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$ (defined also when $a \geq b$).
- (d) $\int_a^c f(x) \, dx = \int_a^b f(x) \, dx + \int_b^c f(x) \, dx$.

Examples:

1. Evaluate $\int_{-2}^5 3 \, dx$, the area under the constant function 3 over the interval $[-2, 5]$.

Solution: The definite integral is the area of a rectangle of height 3 units and width 7, so

$$\int_{-2}^5 3 \, dx = 3 \times 7 = 21.$$

2. Evaluate and interpret the definite integral $\int_5^{-2} 3 \, dx$,

Solution: Moving backwards along the interval $[-2, 5]$ has the effect of measuring the base of a rectangle (appearing in the Riemann sum) as negative. Since the integrand is positive, one expects to produce a negative area. This is confirmed by the following formal manipulation, using the answer to the previous exercise, where we multiply the definite integral by negative one when we swap the terminals:

$$\int_5^{-2} 3 \, dx = - \int_{-2}^5 3 \, dx = -21.$$

3. Evaluate and interpret the definite integral

$$\int_2^2 (f(x) + 3g(x) - 7h(x)) \, dx,$$

where f , g and h are any continuous functions defined on an interval containing $x = 2$.

Solution: The definite integral evaluates to zero, as the interval $[2, 2] = \{2\}$ over which we are integrating is degenerate, consisting of a single point, so the area under the curve vanishes.

4. Evaluate

$$\int_0^1 (2 + 6x^2) \, dx,$$

given that $\int_0^1 x^2 \, dx = \frac{1}{3}$.

Solution: We have

$$\int_0^1 (2 + 6x^2) \, dx = \int_0^1 2 \, dx + 6 \int_0^1 x^2 \, dx = (2 \times 1) + 6 \times \left(\frac{1}{3}\right) = 4.$$

5. Evaluate

$$\int_1^5 (6 - f(x) + 2g(x)) dx ,$$

given that $\int_1^5 f(x) dx = 14$ and $\int_1^5 g(x) dx = -5$.

Solution: We have

$$\begin{aligned} \int_1^5 (6 - f(x) + 2g(x)) dx &= \int_1^5 6 dx - \int_1^5 f(x) dx + 2 \int_1^5 g(x) dx \\ &= (6 \times (5 - 1)) - 14 + 2(-5) = 24 - 14 - 10 = 0 . \end{aligned}$$

6. Find

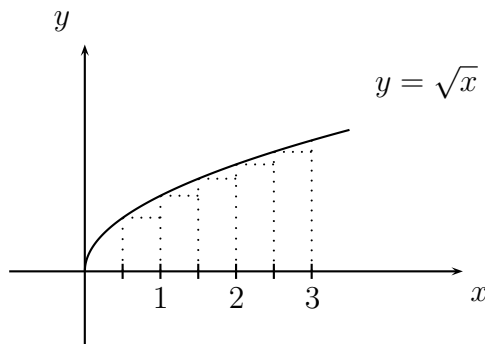
$$\int_1^7 f(x) dx ,$$

given that $\int_3^5 f(x) dx = 2$ and $\int_1^3 f(x) dx = 5$ and $\int_7^5 f(x) dx = -2$.

Solution: We have

$$\begin{aligned} \int_1^7 f(x) dx &= \int_1^3 f(x) dx + \int_3^5 f(x) dx + \int_5^7 f(x) dx \\ &= \int_1^3 f(x) dx + \int_3^5 f(x) dx - \int_7^5 f(x) dx = 5 + 2 - (-2) = 9 . \end{aligned}$$

7. Use a *lower* Riemann sum with 6 subintervals, by finding the areas of the lower rectangles beneath the curve, as indicated in the following diagram, to find a lower bound for the definite integral $\int_0^3 \sqrt{x} dx$, quoting your answer to two decimal places.



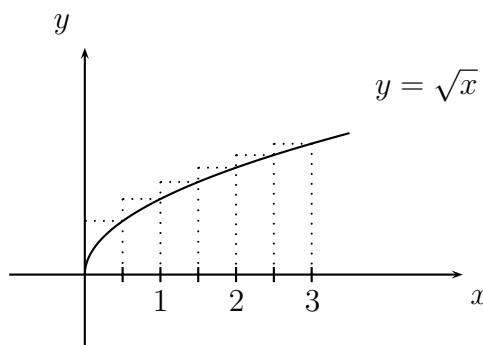
Solution: Each rectangle has width 0.5 and the heights of the lower rectangles are, in succession from left to right,

$$0, \sqrt{0.5}, 1, \sqrt{1.5}, \sqrt{2}, \sqrt{2.5} .$$

Hence the lower Riemann sum is

$$0.5(0 + \sqrt{0.5} + 1 + \sqrt{1.5} + \sqrt{2} + \sqrt{2.5}) \approx 2.96 .$$

8. Use an *upper* Riemann sum with 6 subintervals, by finding the areas of the upper rectangles, as indicated in the following diagram, to find an upper bound for the definite integral $\int_0^3 \sqrt{x} dx$, quoting your answer to two decimal places.



Solution: Each rectangle has width 0.5 and the heights of the rectangles are, in succession from left to right,

$$\sqrt{0.5}, 1, \sqrt{1.5}, \sqrt{2}, \sqrt{2.5}, \sqrt{3}.$$

Hence the upper Riemann sum is

$$0.5(\sqrt{0.5} + 1 + \sqrt{1.5} + \sqrt{2} + \sqrt{2.5} + \sqrt{3}) \approx 3.83.$$

9. Use the average of the lower and upper bounds found in the previous two exercises to estimate $\int_0^3 \sqrt{x} dx$, quoting your answer to one decimal place.

Comment on whether you think this estimate will be an underestimate or an overestimate of the true area under the curve.

Solution: The average of the lower and upper bounds from the previous exercise is

$$\frac{2.96 + 3.83}{2} \approx 3.4,$$

to one decimal place.

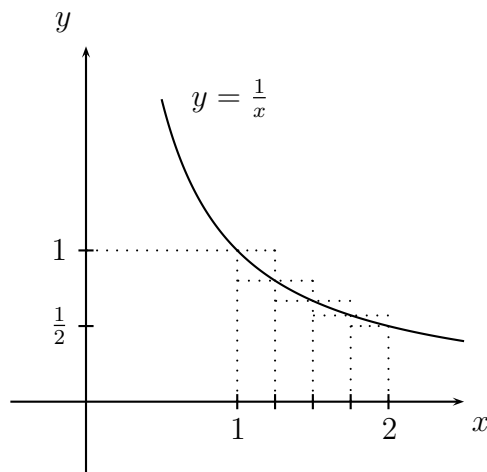
The average of the lower and upper Riemann sums for an increasing function, such as this one, uses areas of trapeziums, so should provide an underestimate of the true area under the curve, since the curve is concave down so sits just above the lines joining endpoints of the subintervals.

Soon you will be able to evaluate this area exactly using the Fundamental Theorem of Calculus. In fact,

$$\int_0^3 \sqrt{x} dx = 2\sqrt{3} \approx 3.5,$$

to one decimal place, which is slightly more than the estimate, confirming our expectations.

10. Use the average of the lower and upper Riemann sums, using 4 subintervals,



to estimate $\int_1^2 \frac{dx}{x}$, rounding off your final answer to three decimal places. Comment on whether you expect your estimate to be an underestimate or overestimate of the true area under the curve.

Solution: The partition of the interval uses the points

$$1, 1.25, 1.5, 1.75, 2.$$

Each subinterval has width 0.25. To form the lower Riemann sum L , we use the right-hand endpoints of the subintervals, and we get the lower bound

$$L = 0.25 \times \left(\frac{1}{1.25} + \frac{1}{1.5} + \frac{1}{1.75} + \frac{1}{2} \right) \approx 0.6345,$$

to four decimal places. To form the upper Riemann sum U , we use the left-hand endpoints of the subintervals, and we get the upper bound

$$U = 0.25 \times \left(\frac{1}{1} + \frac{1}{1.25} + \frac{1}{1.5} + \frac{1}{1.75} \right) \approx 0.7595,$$

to four decimal places. The average of L and U is (keeping everything exact, this time),

$$\frac{L+U}{2} = 0.125 \times \left(\frac{1}{1} + \frac{2}{1.25} + \frac{2}{1.5} + \frac{2}{1.75} + \frac{1}{2} \right) \approx 0.697.$$

We estimate the area under the curve to be about 0.697, and expect this to be a slight overestimate, since the curve is concave up, so the curve sits slightly below lines joining endpoints of subintervals.

Soon you will be able to evaluate this area exactly using the Fundamental Theorem of Calculus. In fact,

$$\int_1^2 \frac{dx}{x} = \ln 2 \approx 0.693,$$

to three decimal places, which is slightly less than the estimate, confirming our expectations.